Assessing Cybersecurity Practices Among Working Journalists

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Abstract: Journalism often requires the need to exchange sensitive and important information from one party to another. Additionally, every year more of these exchanges are transferred to a digital platform. While this may provide better ease for stories to be shared, it also places journalists and their organizations at a vulnerable spot to be monitored. This raises a question on whether journalists are equipped with the proper knowledge and tools to protect themselves and if not, should they be taught better security practices early into their careers. The aim of this study is to understand how journalists view cybersecurity within the scope of their profession. We uncover this through semi-structured style interviews with 5 journalists, exploring how comfortable they are with cybersecurity and how prominent of a thought it is in their lives. In our findings we saw that journalists had a lack of pursuit to enhance their privacy, definitive experiences with cyberattacks, and an understanding of why having strong security would be advantageous. We delve into potential reasons for this lack of concern, pinpointing learned helplessness and an absence of cybersecurity courses offered in universities as a key factor.

CCS Concepts: • **Security and privacy** → **Human and societal aspects of security and privacy**; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: cybersecurity, journalism, privacy, qualitative research, semi-structured interviews, digital security education

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1 Introduction

Reading the news once entailed receiving a paper copy at your doorstep. However, reading newsprint has been drastically replaced by online news media as Journalists adapt to an increasingly digital world. The shift is relatively recent and though journalists adjusted quickly to fast-paced articles and providing coverage as events happen at every moment around the world, an unexpected negative consequence is the cyber risks associated with the profession. Indeed, the convenience of journalists working over their computers, contacting informants over their mobile phones, and news agencies themselves moving their archives to online databases means that they are just as susceptible to targeted cyber-attacks as any other individual or business.

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Journalists tend to carry highly sensitive and private information. These can be emerging stories and investigative work that requires secrecy during collection. Informants and whistle blowers are also a vital way to gather insider information and uncover wrongdoings. The most infamous example is when Edward Snowden shared NSA's massive surveillance with journalists Glenn Greenwald and Laura Poitras. Sharing information to the press poses a lot of risks to those who reach out and may receive backlash for doing so. In more censored states, this can result in physical harm, prison, or even death. If Journalists cannot guarantee a safe channel, or informants are even slightly concerned of the risk, they might hesitate to share information.

Even though the consequences are clear, it is difficult to discern how security tools shape the experience of Journalists. Tor, for example, provides anonymous browsing that common search engines do not, even if using Incognito Mode or InPrivate browser settings. Even though Tor is cited to be used by Journalists for their work, it is also not the most beginner-friendly tool, requiring an in-depth technical understanding of its attack model. It is also subject to slow speeds, increased latency, blocked functionality since most websites use JavaScript, and chances to be deanonymized. There are also many similar security tools that a user must understand well to use properly from VPNs to antivirus software.

Our study has been primarily motivated by how Journalists navigate through the technical details of available tools including Tor, encrypted emails, and secure messaging platforms as well as potential cyber risks and yet still come from a non-tech industry. They are a special population that must guarantee preserving not only their informants' identity but also must consider their personal safety and the information they carry. Though our initial research questions were focused on what activities Journalists may need anonymity for and how much of the underlying technology they felt that they needed to know to use it, we quickly discovered that our participants had minimal to no experience with anonymity tools. Our participants instead shared their thoughts on security and going as far as to detail personal attacks that happened to them. We found their stories as valuable insights from a special population and part of the exploratory process. With this, we shifted our focus to address these questions:

- How do Journalists view cybersecurity within their field of work? What emphasis do they place on security and privacy, including the tools they use?
- Where do Journalists draw their cybersecurity knowledge, such as from their education, careers, or own research?
- What may cause potential low adoption of security tools and practices?

2 Related Work

Majority of the existing work acknowledges that Journalists are especially a point of concern and risk to cyber threats. These works explain that transferring technical knowledge to a primarily non-technical audience is a necessity. Others also point to statistics or publicly known attacks on Journalists and news networks and make inferences with this data. Some also suggest designing an authenticated database or information system that all Journalists can use, regardless of their agency to retrieve verified information. By designing a specialized system, it is possible to gather and share information as opposed to traditional search methods that have potential tracking.

McGregor et al. discovered that Journalists tended to be cyber aware and employ different defensive strategies from tool specific, technological, and non-technical. In her semi-structured interview with 15 Journalists, she found that although technical security expertise varied from low to high, participants would still employ some kind of way to keep themselves, their information and sources safe. Technical defenses included encrypting digital notes, keeping local files as opposed to the cloud, bypassing admin rights on company computers, and even using burner phones. Non-technical strategies were having personal physical safes, handwriting notes, and using code names. Only a few of her participants would use security tools like Password Managers, Tor, or encrypted emails and vary by how regularly they use it.

3 Methodology

To extract the most information about each journalists' stories in the most seamless possible way, we felt that conducting semi-structured interviews would be the ideal format for doing so. We felt that the most beneficial data we would be able to use in this study would be qualitative, so we performed each interview with the goal of generating common themes between each journalist. We held 5 interviews with journalists of a reputable news agency each lasting 10-15 minutes.

3.1 Recruitment

The original plan for recruitment was to gather journalists through both professional networks and word of mouth. However, we were able to conduct multiple interviews within the same news agency in a row due to a connection with one journalist. We were going to send an email that provided each participant with a description about our study and course, but since most of our interviews occurred within the same call this information was explained over the call. Each participant was informed about the handling of the data that they provide for us. We stated that the transcripts of their interview would only be used for the purposes of this class and would not be published anywhere without their permission.

3.2 Interview Structure

An interview script was created so that we could have a foundation for our conversation, which prioritized the most important questions that we wanted to ask each journalist. These questions were tailored towards finding answers to our research questions but were still made open-ended enough to generate follow up topics and create thoughtful discussion regarding each journalists' position on cybersecurity.

Semi-Structured Interview Script:

(1) What does your work primarily consist of?
(2) How much does internet searching and browsing play a role to your work?
(3) Does privacy and security matter for the work you do?
   - What aspects of your work require secrecy and privacy? (Censorship, contacting individuals, investigation, viewing illicit material, your personal safety)
(4) Where do you learn or acquire the knowledge to investigate and collect information in a private matter?
   - Does your education (i.e. university) teach security methods and practices?
(5) To what extent do you feel like you have an adequate understanding of cybersecurity to feel like you are capable of protecting information?
(6) Have you heard of and used Tor? What about Signal, WhatsApp, or Proton Mail in regard to your work?

Questions 1-3 were asked to get a general sense of what each participant's daily work involves and which specific tasks, if any, require any amount of digital security. We wanted to delve into why a journalist might need to be anonymous or be alert towards any vulnerabilities that they have towards potential attackers. Questions 4 and 5 were asked to gauge each participant's knowledge of cybersecurity and awareness of potential threats. Each participant may have different levels of concern over digital attacks, so we wanted to observe how prominent of a thought it was for them on a day-to-day basis. If the participant showed some level of concern regarding cybersecurity, we asked question 6 to see if they used any specific tools or applications that helped them feel more safe working as journalists online. A variety of secure messaging tools were asked about in addition to Tor, in the case that these journalists felt the need to browse potential news stories anonymously.

3.3 Data Analysis

3.3.1 Coding. Each interview was converted into a transcript that was then used for the qualitative coding process. We used inductive coding which means that we started by reviewing our transcripts one by one and derived our codes from them, allowing each unique transcript to potentially generate new topics. Rather than perform rounds of coding, the two of us each generated our own themes and codes based on the transcripts on our own and then came together to compare them. We each re-listened to the recordings and read through the transcripts again to analyze thoroughly and look for common themes found in each interview. During our meeting, if we overlapped on any codes, we would discuss their validity and match them to a corresponding theme. For any codes that each of us came up with that did not overlap, we would try and create a new theme for them that matched our research questions. If we believed they did not fit the overall pursuit of our research, we then discarded them. After this meeting a final code book was generated with all our combined codes in one Excel spreadsheet.

3.3.2 University Course Analysis. Because our participants were well in their careers, and cyber-attacks have only skyrocketed recently, we also investigated whether universities and higher education institutions addressed this phenomenon in the curriculum. Universities are usually the primary way to receive training and prepare students for careers. Universities therefore are responsible in preparing their students and should cater learning to meet what is relevant to the field. By having both semi-structured interviews with professionals in their careers and analysis of curriculum, we can better determine the prevalence of cyber-awareness within the Journalism profession.

We analyzed course lists for Journalism, Media, Communications studies for top-ranking Journalism schools. Using two reputable ranking sources, we gathered the top 50 schools in each one. After removing potential duplicates, the final list contained 94 universities, out of which 9 were selected randomly. 7 institutions are US-based, 1 in Europe, and the other is in Asia. We read through each course and its description for cyber-related themes and key words like "security" "hacking" "encryption" and "privacy." We also kept track of courses that had digital topics, but were not cyber-security focused such as digital media, information technology, or even search engine optimization strategies. These lists were exclusive, so if a course matched cyber themes, it was not also included in the digital technology category. Our process was both LLM-based and manual, where we used the LLM tool to parse key words and categories, but verified the output by manually reading each description as well.

The course lists we analyzed were for Bachelor-level degree programs and did not include Graduate level coursework. We reason that although Graduate programs may have cybersecurity-related specializations, most entry-level Journalism opportunities require bachelor's only, so digital safeguarding knowledge should be available at that level. This increases its exposure to most students since only a select subset of those students will later enroll in Graduate studies.

4 Results

4.1 Key Findings

After reviewing each transcript thoroughly, we came to an agreement on three common themes that the journalists interviewed showcased. The three themes we found were 1. An overall low concern towards digital privacy, 2. A definite understanding of the threat and 3. An awareness of the necessity for cybersecurity. The first theme refers to a lack of attention to protecting one's privacy when carrying out their business online. The focus on this definition is more about the fact that there is no physical action being taken to adding any layer of protection to their digital lives. The second theme is defined as the participant's understanding that there are threats and cyberattacks occurring within their vicinity, workplace, network, etc. The definition of the third theme is that the participant showed awareness that they themselves and other journalists required some means of security in their work life and understand that privacy online is an important factor to protecting oneself.

Code	Code Definition	Examples
Learned Helplessness	Having no control over a situation which leads to giving up on trying to change or fix that issue	Participant #3 believed that security is overall too hard to manage due to its openness. Since cybersecurity is a multi-level threat, between both work and personal life, it's too difficult to stay private. Participant #4 stated that they live with the ongoing monitoring; they are cautious but accepting. The least that they do is not engage with suspicious emails or scams. Participant #1 goes on about their day under the assumption that their phone is tapped.
Lack of Technical Know-How	Does not have a strong understanding on how to incorporate cybersecurity into their everyday workflow	Participant #2 stated that they are "too lazy" to learn about Tor and do not find themselves concerned about searching and tracking. Participants #3 and #4 have never heard about Tor or anonymous browsing.
Increase in Required Protection	The ability to protect one's privacy is becoming increasingly difficult due to advancements in technology and the expanded scope of cybersecurity	Participant #3 talked about the number of open resources available for malicious hackers and that since everything is so accessible now, it makes it very difficult to be private. AI is also a key factor in this since it is now used to make stories "murky" and generate misinformation.

Fig. 1. Qualitative codes to support the claim of low concern towards security from the participants

4.2 Low Action Taken Towards Improving Digital Privacy

For this theme, we determined that there were three main similarities for why journalists may not be taking as much action towards protecting themselves as they might like to be doing. The first code was the idea of learned helplessness. Learned helplessness is the belief that one has little to no control over a situation, leading to them giving up on any attempts to fixing their predicament. This was a thought shared by multiple participants as many accepted that the way they live might just have to be through constant, unwarranted surveillance. One participant stated that the least they could do is not engage with suspicious downloads or emails but all in all live with the ongoing monitoring. Another participant stated that since the cyberattacks can target multiple levels of one's life, like work and personal life, it makes it extremely difficult to manage their privacy. We also found some of our participants using humor to describe their experiences and playing down their experiences, which plays into the psychological harms targeted attacks may have.

Another code that we thought may attribute towards low action is a lack of knowledge in cybersecurity and technology. We defined this code as not having an understanding on how to integrate cybersecurity related tools and notions with the tasks in their work and regular lives that call for them. We found participants to be "too lazy" to learn about tools like "Tor" and are at times not concerned about the fact that their information may be tracked. They do not feel a genuine urgency to understand cybersecurity unless they feel like there is an active threat occurring in their lives.

The last code for this theme is an increase in required protection needed to protect oneself online. Due to the advancements of technology and the rise of AI, the scope of cybersecurity is only growing bigger. This takes its toll on people as well, making it harder for them to believe everything on the internet and harder for them to preserve their privacy. One journalist mentioned the number of resources that are now available for hackers and that since AI is more accessible, it can mud the waters of stories and generate misinformation. This plays an even heavier role for journalists since they have job to maintain the integrity of these stories.

Code	Code Definition	Examples
Company Attacks	Ransomware attacks and company networks being directly targeted by news companies	Participants #2 and #3 both had instances where their company's network and database got hacked into or attempts were made to get into them. For participant #3, data was stolen from the database and used for ransom to release the archives.
Phone Surveillance	Monitoring phone calls between journalists and other parties, wiretapping	Participant #1 was talking over the phone with a colleague who was in an assignment overseas and two minutes of their conversation was played back to them. They will also hear "weird beeps" on their phone. The participant believes that their phones are tapped intentionally because they are journalists. Participant #3 worked and resided primarily overseas. Over their landline, they could hear the last of the conversation being replayed. While covering a news story abroad, they received alerts attempting to change account logins. Their social media accounts were intentionally targeted.

Fig. 2. Qualitative codes showing that participants understood the threat of cyberattacks

4.3 A Definite Understanding of the Threat

To show that the participants have a definite grasp of the threat, we found two common examples of attacks that the journalists had experienced. The first example of a frequent attack was attacks on the company, whether that be on the network or the company databases. One participant recalled an instance where their network was compromised and caused a complete shutdown, so no work was able to be done. Another participant stated that data was stolen from the company database and used for ransom, attempting to exchange the archives for money.

The second type of attack that was most notable from their experiences was phone surveillance. This would be the act of monitoring phone calls and recording conversations more notably known as wiretapping. We talked with two participants who both had instances where they were having a conversation on the phone and could hear a playback of the last minutes of their talk. One participant believed that their phone was intentionally targeted because they are a journalist. Phone surveillance can also be done through accessing social media accounts which was seen in Participant 3. While working on a news story abroad, they found that once they were in that location they received alerts that there were attempts to change their account's logins.

4.4 An Awareness of the Necessity for Cybersecurity

Code	Code Definition	Examples
Encrypted Messaging	Incorporating a convenient method of privacy and connection by using encrypted messages	All participants use Signal, which is an end-to-end encrypted messaging app focused on privacy.
Superficial Concerns	Rather than expressing the direct harm cyberattacks have towards the individual, the danger is said to be placed on the entire network	Participant #2 would talk about the entire network being attacked in their company rather than the impact it had on them individually. Participant #3, when mentioning an attack on the company database, expressed what repercussions it had on the company but not on them directly.

Fig. 3. Qualitative codes showing that participants are aware that cybersecurity is needed

We found that even though the participants might not be making conscious efforts each day of their lives to be private, they are still taking steps to being secure in their profession and are aware that cybersecurity is needed. The first commonality that we noted was that all participants use encrypted messaging through Signal, an end-to-end encrypted messaging app. Signal is a more usable and convenient method of having private conversations and the journalists use it to message others in their work.

We also discovered that although the participants were expressing a recognition for security needs, their concerns were rather abstract and focused on the impact the attack had on the company or network not on themselves individually. Participant 3, who mentioned the attack that occurred on the company database, expressed the consequences it had on the company but not exactly how it affected himself and whether he felt any kind of invasion of privacy.

4.5 Minimal to No Cybersecurity Courses

Our analysis of 9 highly praised institutions for Journalism, Media, and Communication studies resulted in a collection of 505 courses. Each institution offered an average of 56.11 courses and a median of 48. After reading each course description, we determined only 5 out of the 505 courses available were cybersecurity related, covering topics like Edward Snowden's press release, encryption, cyber policy, and AI use for social media surveillance. 2 of these courses were from the same institution. Only 1 course out the 5 found was exemplary as it directly addresses the issue of protecting digital privacy that students can practically use. The course description is included below:

We all use computers, smartphones, & the internet, but most of us don't know much about securing our devices, our personal data, or even our identity. There's a lot to worry about-viruses, phishing, surveillance, tracking, & much more-but what should really concern you? How can you tell if you or your devices are vulnerable? How can you safeguard yourself? This course, meant for the average technology user, will answer these questions and many more. Topics covered include passwords, spyware, messaging & email, encryption, testing for vulnerabilities, HTTPS, and backup.

Table 1. Summary of Course Offerings by Institution

ID	Location	Courses	Cyber	Digital
1	US	41	2	12
2	US	40	1	17
3	US	55	1	17
4	US	62	0	22
5	US	17	0	7
6	US	55	0	20
7	US	48	0	11
8	Europe	36	0	0
9	Asia	151	1	56
Total		**505**	**5**	**162**

4.6 Trending and Widely Available Digital and Technology Courses

Even though cybersecurity was a rarely addressed topic with most universities not offering any specific training, we found that institutions were well-tuned in ensuring students had general digital and technical know-how. Overall, we found 162 digital courses that ranged from digital media, writing stories for specific platforms, search engine optimization, to AI/AR/VR and drone technologies. Having such varied courses showcases that higher education institutions acknowledge that their students can encounter these topics within their careers. On average, each university offered 18 digital and technological courses with a median of 17.

5 Discussion

5.1 Increased Acceptance of Threat

It was evident that many of our participants had experienced at least some form of a legitimate cyber threat (network hacking, phone surveillance, etc.) in their lives. We believe that these instances combined with the increasing amount of information available for hackers, make it difficult for these journalists to believe that improvements to their data protection will be made. As these situations continuously arise, they will begin to tolerate this ongoing monitoring and lose confidence that their actions will create any beneficial change to their security.

5.2 Uncertainty Over Responsibility

Since we found participants to describe the attacks as targeting the entire network rather than themselves, we found that this may lead to confusion regarding whose responsibility it is to become more knowledgeable about cybersecurity practices. Although there may be journalists in the company who are tech-savvy, this does not discount the ones who still have knowledge gaps. There may be an expectation for the more well informed to be the ones to fix any security related issues which places less of a burden on the journalists who are not as educated in that realm. However, this can be dangerous once a journalist is targeted individually and does not have the proper background to protect themselves.

It is noted that certain security related tools do not integrate well with the communication necessary for journalists, which is why Signal seems to be adopted by many. However, while decisions like using Signal to communicate and exchange information show that the participants are beginning to incorporate more security into their work, we believe that there are more malicious methods of hacking like spyware and malware that won't be able to be prevented just by encrypting messages. Additionally, even though some of the journalists we interviewed are aware of the potential for more severe attacks, we believe that a benefit could be made by having more agency over one's information rather than relying on being taught to protect it.

5.3 Role Alignment

Within Journalism, there are Investigative specialties and Mass Media specialties. Tying in with responsibility being diffused to more tech-savvy individuals or across the company, we found that Mass Media Journalists tend to think cyber security was not needed for their line of work. With Investigate Journalism, the Mass Media group reasoned, there may be more tech-savvy individuals who take cyber security seriously. Comparing to Susan McGregor's work, her pool of participants definitely employed multiple defensive strategies. Though most her participants did not use Tor or other advanced security tools or strategies, they generally seemed to be more defensive than our pool of participants. This could be because McGregor's population had more investigative journalists present while ours leaned toward the mass media side.

5.4 Noticeable Shift on Source Communication

We found that all Journalists use Signal. Our participants used it unanimously with one going as far to say that it is an industry practice now. This shift is noticeable and interesting because in McGregor's work, Journalists tended to accommodate to what their source felt comfortable using even if they personally did not prefer it such an unencrypted email. The shift not only highlights a brand-new tool introduced than what was used before, like encrypted emails or anonymous platforms where authenticity also plays a role, but also that Journalists specifically outline to reach them out through Signal. They now set an expectation that they are reachable, but sources are expected to contact them with information through specific channels only and this being Signal. Compared to anonymous platforms like Tor, Signal is widely-accessible and easy to use which may explain the shift from previously vetted platforms to new ones.

5.5 Lack of Cyber-preparedness in both Career and Study

Our interviews with professionals in the field, with some working for decades, may have indicated that cyber threats only appeared while on the job and could not have received formal training or preparedness from their education. However, what we determined was that even new Journalists and prospective students that are currently studying also have very limited training on cyber threats. Although universities prepare their students with overall digital technological trends with AI being the most prominent, students do not perceive the risks associated with the job to their personal safety or to their sources. Additionally, most of the cyber and digital classes were electives or optional. The 1 exemplary cyber class was 1 credit, which may be intentionally designed to have many students enroll in a low commitment class. However, in general, we can infer that it is up to the students' discretion whether to enroll in digital or cyber classes and the university did not make these mandatory. The student needs to deem how important cyber and digital values are to them. What we found was overall not enough training from both in the profession through their agencies or from higher education.

5.6 Level of Malware and Spyware is Unknown

With limited inference on what malware or spyware could be affecting individual participants, we are unsure what bad actors use for their targeted attacks. Our participants vouch for Signal which ensures a secure communication medium, but it does not prevent specific malware or spyware from causing harm on their personal devices. It is unclear who could be behind such an attack and whether it is the same individuals or groups. Even among participant to participant, it is entirely possible that each one has a different or set of malwares, or there may be even multiple sets of attackers. This leaves many unsatisfactory questions since the goal of this research space is to help individuals which is difficult without full knowledge of the extent of the malware.

6 Future Work

A possible direction for future research could be a more generalized qualitative study done through a survey, administered to both journalists and current journalism majors. The journalists can be in various fields (investigative, mass media, etc.) to distinguish if any fields have more knowledge in cybersecurity than others. The survey would ask questions regarding what tools they use if any and how they believe the journalism field could benefit from advanced cybersecurity. It would be beneficial to measure how informed journalism majors are in privacy and tech, specifically ones that are pursuing a career in investigative journalism.

Another future project would be to raise awareness to serious cyber-attacks, like spyware, that journalists can fall victim to that are not as easy to prevent with more usable security software. It would be a significant project since we believe that we could enhance the comfortability of journalists with cyber tools and make them feel like their data and privacy is within their own control.

We also believe that another avenue of future work is to analyze the malware on targeted victims' devices through digital forensics. This can lead to insights on what attackers use, which can give very specific advice to victims and potential preventive measures to such malware. It would require close collaboration with affected Journalists with the goal being to bring back security and privacy into their lives.

7 Limitations

Our participants would detail the malware on their devices, but due to descriptions of their experience which has very insightful, we could not determine the exact potential spyware used. Figuring out the specific malware on their devices would perhaps require forensics that was not in the scope of this study. This limits us on how to give better, specific recommendations and protective measures that we could have shared to our participants and Journalists to recover from such attacks or prevent them in the first place.

When reviewing course lists, each institution has their individual organization schemes. We standardized the review by limiting course lists to only those mandatory to the major and within the Journalism, Media, and Communication fields. For European or Asian universities that use different measurement systems, it could be possible that one unit in the European or Asian university equates two courses in the United States. This is especially true for year-long courses in European institutions, which were then divided into two counts to match American schools. Within the credit system, some courses are for 1 credit, while others are for 3 or 4. Therefore, there wasn't a weighted sum applied to the course counts that shows the significance that the university applied to that specific course.

We also could not consider the entire Journalism major experience, where universities usually require outside-of-major requirements like technology, Math, or Sciences and that students have outside-of-class internships and work-related experience. These can be opportunities to learn about cyber security which we could not reasonably account for since it depends on the individual student. This can also be an interesting avenue for future work.

8 Conclusion

Since journalists can handle controversial subjects and sensitive stories that could place both them and the source of that information at risk, we believe that they are one of the most appropriate professions that should require an understanding of cybersecurity tactics. To get a better understanding of how journalists can better integrate these tools into their work, we needed to discover what cyber threats they have been victims of and how knowledgeable they were in the field of cybersecurity. We conducted 5 semi-structured interviews, asking questions that were directed towards first finding out each journalist's comfortability with cybersecurity and based off their responses, uncovering how far of an attempt they make to maintain privacy in their day-to-day lives.

The journalist's responses showed a low level of concern towards cyber-related threats and our analysis, done using qualitative coding, provided potential intuition to why they are not acting to fix their privacy. Participants have been exposed to numerous cyber-attacks in which they have no control over, diminishing their confidence in fixing the issue and leading to an acceptance of the problem. The reasoning for existing knowledge gaps in our participants was also explored through the availability of cybersecurity courses in universities. We found an overall deficiency in digital safety classes offered and discovered only 1 ideal course offered that addresses how to be private online.

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A Interview Questions

Semi-Structured Interview Script:

(1) What does your work primarily consist of?
(2) How much does internet searching and browsing play a role to your work?
(3) Does privacy and security matter for the work you do?
   - What aspects of your work require secrecy and privacy? (Censorship, contacting individuals, investigation, viewing illicit material, your personal safety)
(4) Where do you learn or acquire the knowledge to investigate and collect information in a private matter?
   - Does your education (i.e. university) teach security methods and practices?
(5) To what extent do you feel like you have an adequate understanding of cybersecurity to feel like you are capable of protecting information?
(6) Have you heard of and used Tor? What about Signal, WhatsApp, or Proton Mail in regard to your work?

B Codebook

Theme	Code	Code Definition	Examples
Low Concern Towards Digital Privacy	Learned Helplessness	Having no control over a situation which leads to giving up on trying to change or fix that issue	Participant #3 believed that security is overall too hard to manage due to its openess. Since cybersecurity is multi-level threat, between both work and person life, its too difficult to stay private Participant #4 stated that they live with the ongoing monitoring, they are cautious but accepting. The least that they do is not engage with suspicious emails or downloads Participant #1 goes on about their day under the assumption that their phone is tapped
	Lack of Technical Know-How	Does not have a strong understanding on how to incorporate cybersecurity into their everyday workflow	Participant #2 stated that they are "too lazy" to learn about Tor and does not find themselves concerned about searching and tracking Participants #3 and #4 have never heard about Tor or anonymous browsing
	Increase in Required Protection	The ability to protect oneselve's privacy is becoming increasingly difficult due to advancements in technology and the grand scope of cybersecurity	Participant #3 talked about the number of open resources available for malicious hackers and that since everything is so accessible now, it makes it very difficult to be private. AI is also a key factor in this since it is now used to make stories "murky" and generate misinformation.
Definite Understanding of the Threat	Company Attacks	Cyberattacks and ransom made directly towards news companies	Participants #2 and #3 both had instances where their company's network and database got hacked into or attempts were made to get into them. For participant #3, data was stolen from the database and used for ransom to release the archives.
	Phone Surveillance	Monitoring phone calls between journalists and other parties, wiretapping.	Participant #1 was talking with a colleague who was in Jerusalem over the phone and two minutes of their conversation was played back. They will also hear "weird beeps" on her phone. Believes that their phones are tapped intentionally due to the fact that they are journalists. Participant #3 worked in Belfast and had a phone call where he could hear the last 10 seconds of phone being replayed. In Jerusalem, was covering news story and got email alerts saying that account passwords trying to be changed. All of his social media account where he was posting news were intentionally being targeted.
Aware of Cybersecurity Necessity	Encrypted Messaging	Incorporating a convenient method of privacy and protection by sending encrypted messages	Participants #2 , #4, #5 all use Signal which is an end-to-end encrypted messaging app focused on privacy.
	Superficial Concerns	Rather than expressing the direct harm cyberattacks have towards the individual, the danger is said to be placed on the entire network	Participant #2 would talk about the entire network being attacked in their company rather than the impact it had on her individually. Participant #3, when mentioning an attack on the company database, expressed what repercussions it had on the company (took precautions, VPN InCognito modes, antivirus modes) but not on them directly

Fig. 4. Qualitative codebook used for thematic analysis